

sider a case in which the probability of rain is $1/2$ and $Z = \{z_1, z_2\}$, where $z_1 = \text{“having an umbrella”}$ and $z_2 = \text{“not having an umbrella”}$. A “lottery” in which you have z_1 if it is raining and z_2 if it is not, should not be considered equivalent to the “lottery” in which you have z_1 if it is not raining and z_2 if it is. Thus, we have to be careful not to apply the model in contexts where the attitude toward the consequence depends on the event realized in each possible contingency.

Preferences

Let us think about examples of “sound” preferences over a space $L(Z)$. Following are some examples:

- *Preference for uniformity*: The decision maker prefers the lottery that is less disperse where dispersion is measured by $\sum_z (p(z) - 1/|Z|)^2$.
- *Preference for most likelihood*: The decision maker prefers p to q if $\max_z p(z)$ is greater than $\max_z q(z)$.
- *The size of the support*: The decision maker evaluates each lottery by the number of prizes that can be realized with positive probability, that is, by the size of the support of the lottery, $\text{supp}(p) = \{z | p(z) > 0\}$. He prefers a lottery p over a lottery q if $|\text{supp}(p)| \leq |\text{supp}(q)|$.

These three examples are degenerate in the sense that the preferences ignored the consequences and were dependent on the probability vectors alone. In the following examples, the preferences involve evaluation of the prizes as well.

- *Increasing the probability of a “good” outcome*: The set Z is partitioned into two disjoint sets G and B (good and bad), and between two lotteries the decision maker prefers the lottery that yields “good” prizes with higher probability.
- *The worst case*: The decision maker evaluates lotteries by the *worst possible* case. He attaches a number $v(z)$ to each prize z and $p \succsim q$ if $\min\{v(z) | p(z) > 0\} \geq \min\{v(z) | q(z) > 0\}$. This criterion is often used in computer science, where one algorithm is preferred to another if it functions better in the worst case independently of the likelihood of the worst case occurring.
- *Comparing the most likely prize*: The decision maker considers the prize in each lottery that is most likely (breaking ties in some

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- $ZA: N.O.$ $Z = \{z_1, z_2, z_3\}$, $z_1 \succ z_2 \succ z_3$
 $\pi^1 = (0.6, 0.4, 0)$, $\pi^2 = (0.4, 0.6, 0)$, $\pi^3 = (0, 0, 1)$
 $\pi^1 \succ \pi^2$, But $0.5\pi^1 + 0.5\pi^3 \sim 0.5\pi^2 + 0.5\pi^3$

for equally likely
outcome, look at the
better one.

$C = \{x, y\}, x \succ y, \pi^1 = (1, 0), \pi^2 = (0, 1) \Rightarrow \pi^3 =$
 $\{\alpha \in [0, 1]: \pi^2 \succeq \alpha \pi^1 + (1 - \alpha) \pi^3\} = [0, 0.5).$

arbitrary way) and compares two lotteries according to a basic preference relation over Z .

- *Lexicographic preferences*: The prizes are ordered $z_1, \dots, z_K$, and the lottery p is preferred to q if $(p(z_1), \dots, p(z_K)) \geq_L (q(z_1), \dots, q(z_K))$.
- *Expected utility*: A number $v(z)$ is attached to each prize, and a lottery p is evaluated according to its expected v , that is, according to $\sum_z p(z)v(z)$. Thus,

$$p \succsim q \text{ if } U(p) = \sum_{z \in Z} p(z)v(z) \geq U(q) = \sum_{z \in Z} q(z)v(z).$$

Note that the above examples constitute ingredients that could be combined in various ways to form an even richer class of examples. For example, one preference can be employed as long as it is “decisive”, and a second preference can be used to break ties when it is not.

The richness of examples calls for the classification of preference relations over lotteries and the study of properties that these relations satisfy. The methodology we follow is to formally state general principles (axioms) that may apply to preferences over the space of lotteries. Each axiom carries with it a consistency requirement or involves a procedural aspect of decision making. When a set of axioms characterizes a family of preferences, we will consider the set of axioms as justification for focusing on that specific family.

von Neumann and Morgenstern Axiomatization

The version of the von Neumann and Morgenstern axiomatization presented here uses two axioms, the independence and continuity axioms.

The Independence Axiom

In order to state the first axiom, we require an additional concept, called *Compound lotteries* (fig. 7.1): Given a K -tuple of lotteries $(p^k)_{k=1, \dots, K}$ and a K -tuple of nonnegative numbers $(\alpha_k)_{k=1, \dots, K}$ that sum up to 1, define $\oplus_{k=1}^K \alpha_k p^k$ to be the lottery for which $(\oplus_{k=1}^K \alpha_k p^k)(z) = \sum_{k=1}^K \alpha_k p^k(z)$. Verify that $\oplus_{k=1}^K \alpha_k p^k$ is indeed a lottery. When only two lotteries p^1 and p^2 are involved, we use the notation $\alpha_1 p^1 \oplus (1 - \alpha_1) p^2$.

We think of $\oplus_{k=1}^K \alpha_k p^k$ as a compound lottery with the following two stages:

Stage 1: It is randomly determined which of the lotteries $p^1, \dots, p^K$ is realized; α_k is the probability that p^k is realized.